Chapter 4. Differentiable Functions

Derivatives, mean value theorems, L'Hôpital's rule, Darboux's theorem, Taylor's theorem

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4.1 Differentiability

Motivation. The derivative is the best linear approximation; its rules follow from $f(x+h) = f(x) + f'(x)h + o(h)$.

Let I be an interval. $f : I \to \mathbb{R}$ is **differentiable** at $x_0 \in I$ if $f'(x_0) = \lim_{h\to 0} \frac{f(x_0+h)-f(x_0)}{h}$ exists (one-sided at an end point).

Proposition 4.1 (4.1.1)

For $f : (a, b) \to \mathbb{R}$ and $x_0 \in (a, b)$ the following are equivalent:
(a) f is differentiable at x_0;
(b) there are $\lambda \in \mathbb{R}$ and a function ϕ with $f(x_0+h)-f(x_0) = \lambda h+|h|\phi(h)$ and $\lim_{h\to 0} \phi(h) = 0$.
Then $\lambda = f'(x_0)$.

Proof. (a)⇒(b): put $\lambda = f'(x_0)$ and $\phi(h) = (f(x_0 + h) - f(x_0) - \lambda h)/|h|$ for $h \neq 0$, $\phi(0) = 0$.
(b)⇒(a): $\frac{f(x_0+h)-f(x_0)}{h} = \lambda \pm \phi(h) \to \lambda$. ■

This linear-approximation form generalizes to several variables.

Proposition 4.2 (4.1.2)

If f is differentiable at x_0, it is continuous at x_0.

Proof. $f(x_0 + h) - f(x_0) = \lambda h + |h|\phi(h) \to 0$. ■

Proposition 4.3 (4.1.3)

If f, g are differentiable at x_0, so are $f+g$, fg, and f/g (if $g(x_0) \neq 0$), with $(f+g)' = f'+g'$, $(fg)' = f'g + fg'$, $(f/g)' = (f'g - fg')/g^2$ at x_0.

Proof. For the product, $\frac{f(x_0+h)g(x_0+h)-f(x_0)g(x_0)}{h} = \frac{f(x_0+h)-f(x_0)}{h} g(x_0+h) + f(x_0)\frac{g(x_0+h)-g(x_0)}{h}$, and $g(x_0 + h) \to g(x_0)$ by Proposition 4.2. For the quotient, first $\frac{1/g(x_0+h)-1/g(x_0)}{h} = -\frac{g(x_0+h)-g(x_0)}{h} \cdot \frac{1}{g(x_0+h)g(x_0)} \to -g'/g^2$, then use the product rule. ■

Proposition 4.4 (4.1.4, chain rule)

If $f : I \to J$ is differentiable at x_0 and $g : J \to \mathbb{R}$ is differentiable at $y_0 = f(x_0)$, then $g \circ f$ is differentiable at x_0 and $(g \circ f)'(x_0) = g'(y_0)f'(x_0)$.

Proof. Write $g(y_0 + k) - g(y_0) = k(g'(y_0) + \psi(k))$ with $\psi(k) \to 0 = \psi(0)$. With $k = f(x_0 + h) - f(x_0) \to 0$,

$$\frac{g(f(x_0 + h)) - g(f(x_0))}{h} = \frac{k}{h}\,(g'(y_0) + \psi(k)) \to f'(x_0)g'(y_0).$$

■

Example 4.1

$f(x) = x^2 \sin(1/x)$, $f(0) = 0$, is differentiable everywhere with $f'(0) = \lim h \sin(1/h) = 0$, but $f'(x) = 2x \sin(1/x) - \cos(1/x)$ is not continuous at 0. $|x|$ is continuous but not differentiable at 0.

4.2 Mean value theorems

Motivation. The mean value theorem turns local derivative information into global conclusions.

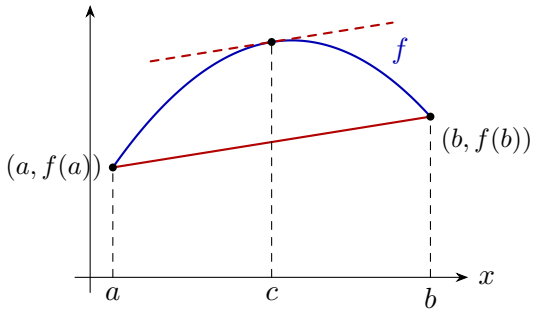


Figure. *Mean value theorem: the tangent at c (dashed) is parallel to the secant.*

If f has a local maximum or minimum at an interior point c and is differentiable there, then $f'(c) = 0$: the difference quotients have opposite signs on the two sides.

Lemma 4.5 (4.2.1, Rolle)

If $f : [a, b] \to \mathbb{R}$ is continuous on $[a, b]$, differentiable on (a, b), and $f(a) = f(b)$, then $f'(c) = 0$ for some $c \in (a, b)$.

Proof. If f is constant, any c works. Otherwise f attains a maximum or minimum (Corollary 3.9) at a point $c \in (a, b)$ where $f(c) \neq f(a)$, and $f'(c) = 0$. ∎

Theorem 4.6 (4.2.2, Cauchy mean value theorem)

If $f, g : [a, b] \to \mathbb{R}$ are continuous on $[a, b]$ and differentiable on (a, b), there is $c \in (a, b)$ with $f'(c)[g(b) - g(a)] = g'(c)[f(b) - f(a)]$.

Proof. Apply Rolle to $h(x) = f(x)[g(b) - g(a)] - g(x)[f(b) - f(a)]$, for which $h(a) = h(b) = f(a)g(b) - g(a)f(b)$. ∎

Corollary 4.7 (4.2.3, mean value theorem)

If f is continuous on $[a, b]$ and differentiable on (a, b), then $f(b) - f(a) = f'(c)(b - a)$ for some $c \in (a, b)$.

Proof. Take $g(x) = x$. ∎

Corollary 4.8 (4.2.4)

For differentiable $f : (a, b) \to \mathbb{R}$: f is increasing iff $f' \geq 0$; constant iff $f' = 0$; decreasing iff $f' \leq 0$. If $f' > 0$, f is strictly increasing.

Proof. If $f' \geq 0$ and $x < y$, $f(y) - f(x) = f'(c)(y - x) \geq 0$. Conversely, if f increases, its difference quotients are ≥ 0, so $f' \geq 0$. ∎

Theorem 4.9 (4.2.5, L'Hôpital's rule)

Let f, g be differentiable near x_0 (except possibly at x_0) with $g' \neq 0$ there, $\lim_{x\to x_0} f(x) = \lim_{x\to x_0} g(x) = 0$, and $\lim_{x\to x_0} f'(x)/g'(x) = L$. Then $\lim_{x\to x_0} f(x)/g(x) = L$. The same holds for one-sided limits, for $x \to \pm\infty$, and when $g(x) \to \pm\infty$.

Proof. Set $f(x_0) = g(x_0) = 0$. Near x_0, $g(x) \neq 0$ (Rolle, $g' \neq 0$), and Theorem 4.6 gives $f(x)/g(x) = f'(c_x)/g'(c_x)$ with c_x between x_0 and x, so $c_x \to x_0$. For $x \to \infty$ put $x = 1/t$.

If $g \to \infty$ as $x \to x_0^+$: choose δ with $|f'/g' - L| < \epsilon$ on $(x_0, x_0 + \delta)$, fix y there; for $x_0 < x < y$, $\frac{f(x)-f(y)}{g(x)-g(y)} = \frac{f'(c)}{g'(c)}$ and

$$\frac{f(x)}{g(x)} = \frac{f(x) - f(y)}{g(x) - g(y)} \cdot \frac{1 - g(y)/g(x)}{1} + \frac{f(y)}{g(x)},$$

where $g(y)/g(x) \to 0$ and $f(y)/g(x) \to 0$ as $x \to x_0^+$; so $\limsup|f/g - L| \leq \epsilon$. ■

Theorem 4.10 (4.2.6, Darboux)

If $F : [a, b] \to \mathbb{R}$ is differentiable and $F'(a) < \alpha < F'(b)$, then $F'(c) = \alpha$ for some $c \in (a, b)$.

Proof. $G(x) = F(x) - \alpha x$ has $G'(a) < 0 < G'(b)$, so $G(x) < G(a)$ for x slightly larger than a and $G(x) < G(b)$ for x slightly smaller than b. Hence the minimum of G on $[a, b]$ is attained at an interior point c, where $G'(c) = 0$. ■

Corollary 4.11 (4.2.7)

If $f : I \to \mathbb{R}$ has a primitive on the interval I, then $f(J)$ is an interval for every interval $J \subset I$.

Proof. Apply Theorem 4.10 to the primitive on subintervals of J. ■

So a function with a jump discontinuity, such as the sign function, has no primitive; derivatives can be discontinuous (Example 4.1) but have the intermediate value property.

4.3 Taylor's theorem

Motivation. Taylor's theorem bounds the error of approximating f by its Taylor polynomial.

For f with n derivatives at a, let $T_n^f(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!}(x - a)^k$ be its Taylor polynomial at a.

Theorem 4.12 (4.3.1, generalized mean value theorem)

Let I be the closed interval with end points $a \neq b$, and $f, g : I \to \mathbb{R}$ with $f^{(n)}, g^{(n)}$ continuous on I and differentiable on $\operatorname{Int} I$. For each $x \in I$, $x \neq a$, there is c strictly between a and x with

$$[f(x) - T_n^f(x)]g^{(n+1)}(c) = [g(x) - T_n^g(x)]f^{(n+1)}(c).$$

Proof. Fix x and let $F(t) = \sum_{k=0}^n \frac{f^{(k)}(t)}{k!}(x - t)^k$, $G(t)$ likewise for g, for t between a and x. In $F'(t)$ the terms telescope:

$$F'(t) = \sum_{k=0}^{n} \frac{f^{(k+1)}(t)}{k!}(x - t)^k - \sum_{k=1}^{n} \frac{f^{(k)}(t)}{(k-1)!}(x - t)^{k-1} = \frac{f^{(n+1)}(t)}{n!}(x - t)^n,$$

and similarly for G. Also $F(x) = f(x)$ and $F(a) = T_n^f(x)$. Theorem 4.6 gives c with $[F(x) - F(a)]G'(c) = [G(x) - G(a)]F'(c)$; cancel the nonzero factor $(x-c)^n/n!$. ■

Corollary 4.13 (4.3.2, Taylor's theorem)

Under the hypotheses on f, for each $x \in I$ there is c between a and x with

$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k + \frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1}.$$

Proof. Apply Theorem 4.12 with $g(t) = (t-a)^{n+1}$: $T_n^g = 0$ and $g^{(n+1)} = (n+1)!$. ■

The last term is the **remainder**. If f is C^∞, the **Taylor series** of f at a is $\sum_k \frac{f^{(k)}(a)}{k!}(x-a)^k$; it may fail to converge to f: $f(x) = e^{-1/x^2}$, $f(0) = 0$, has all derivatives 0 at 0.

Proposition 4.14 (4.3.3)

Let f be C^∞ on I, $a, x \in I$, and J the closed interval between them. If $\{|f^{(n)}(t)| : t \in J, n \geq 0\}$ is bounded, the Taylor series of f at a converges to $f(x)$.

Proof. With M a bound, the remainder is at most $M|x-a|^{n+1}/(n+1)! \to 0$ (the terms of the convergent series $\sum r^n/n!$ tend to 0). ■

Example 4.2

e^x, $\sin x$, $\cos x$ satisfy the hypothesis on every bounded interval, so $e^x = \sum x^n/n!$, $\sin x = \sum(-1)^n x^{2n+1}/(2n+1)!$, $\cos x = \sum(-1)^n x^{2n}/(2n)!$ for all real x. Taylor's theorem also gives the second derivative test: if $f'(a) = 0$ and $f''(a) > 0$, f has a strict local minimum at a.

4.4 Selected exercises

Motivation. Four book exercises using the mean value theorem, Rolle and L'Hôpital.

Exercise 4.4.8 (A Hölder condition of order > 1)

Let I be an interval, $M > 0$, $\alpha > 1$, and $f : I \to \mathbb{R}$ with $|f(x) - f(y)| \leq M|x-y|^\alpha$ for all $x, y \in I$. Show that f is constant. (The book's case is $M = 1$, $\alpha = 2$, $I = \mathbb{R}$.)

Solution.

Step 1. For $x \in I$ and $y \in I$, $y \neq x$,

$$\left|\frac{f(y) - f(x)}{y - x}\right| \leq M|y-x|^{\alpha-1} \longrightarrow 0 \quad (y \to x),$$

since $\alpha - 1 > 0$. So f is differentiable on I with $f' \equiv 0$ (one-sided at end points).

Step 2. For $a < b$ in I, $f(b) - f(a) = f'(c)(b-a) = 0$ (Proposition 4.2, Corollary 4.7).

It fails for $\alpha = 1$ ($f(x) = x$) and $\alpha = \frac{1}{2}$ ($\sqrt{x}$ on $[0, \infty)$, Example 3.3). ■

Exercise 4.4.9 (Limits of the derivative)

Let I be an open interval, $c \in I$, and $f : I \to \mathbb{R}$ continuous at c and differentiable on $I - \{c\}$. Show that if $\lim_{x\to c} f'(x) = K$ exists, then f is differentiable at c and $f'(c) = K$.

Solution.

Step 1. Let $\epsilon > 0$. Choose $\eta > 0$ with $[c-\eta, c+\eta] \subset I$ and $|f'(z) - K| < \epsilon$ for $0 < |z - c| < \eta$.

Step 2. For $0 < h < \eta$, f is continuous on $[c, c+h]$ and differentiable inside, so by Corollary 4.7 some $z \in (c, c+h)$ has

$$\frac{f(c+h) - f(c)}{h} = f'(z), \qquad \text{so} \qquad \left|\frac{f(c+h) - f(c)}{h} - K\right| < \epsilon.$$

Step 3. Likewise for $-\eta < h < 0$; so $f'(c) = K$.

Hence derivatives have no removable or jump discontinuities; a discontinuous derivative (Example 4.1) has no limit there. ■

Exercise 4.4.12 (Roots of $\frac{d^n}{dx^n}(x^2 - x)^n$)

Let $n \geq 1$ and $f(x) = (x^2 - x)^n = x^n(x-1)^n$. Show that $f^{(n)}$ has exactly n distinct roots, all in $(0, 1)$.

Solution.

Step 1. For $0 \leq k \leq n$, $f^{(k)}(x) = x^{n-k}h_k(x)$ for some polynomial h_k. This holds for $k = 0$ with $h_0 = (x-1)^n$; if it holds for $k < n$, then with $m = n - k \geq 1$,

$$f^{(k+1)}(x) = mx^{m-1}h_k(x) + x^m h_k'(x) = x^{m-1}(mh_k(x) + xh_k'(x)).$$

Hence $f^{(k)}(0) = 0$ for $0 \leq k \leq n-1$. Writing $f(x) = (x-1)^n x^n$, the same argument with $x - 1$ in place of x gives $f^{(k)}(1) = 0$ for $0 \leq k \leq n-1$.

Step 2. Claim: $f^{(k)}$ $(1 \leq k \leq n)$ has $\geq k$ distinct roots in $(0, 1)$. $k = 1$: Rolle (Lemma 4.5) on $[0, 1]$. If $k \leq n-1$ and $0 < \alpha_1 < \cdots < \alpha_k < 1$ are roots of $f^{(k)}$, by Step 1 $f^{(k)}$ vanishes at the $k+2$ points

$$0 < \alpha_1 < \cdots < \alpha_k < 1.$$

and Rolle on the $k+1$ disjoint gaps gives $k+1$ distinct roots of $f^{(k+1)}$.

Step 3. $f^{(n)}$ has degree n (leading coefficient $(2n)!/n!$), hence at most n roots; with Step 2, exactly n, all in $(0, 1)$.

Up to a constant and $x = (1+t)/2$, $f^{(n)}$ is the n-th Legendre polynomial. ■

Exercise 4.4.13 (Symmetric second difference quotient)

(a) Let f be differentiable on an open interval $I \ni c$ and suppose $f''(c)$ exists. Show that

$$\lim_{h\to 0} \frac{f(c+h) + f(c-h) - 2f(c)}{h^2} = f''(c).$$

(b) Show that the limit in (a) may exist although $f''(c)$ does not.

Solution.

(a)

Step 1. $N(h) = f(c+h) + f(c-h) - 2f(c)$ and $D(h) = h^2$ are differentiable, tend to 0, and $D'(h) = 2h \neq 0$.

Step 2.

$$\frac{N'(h)}{D'(h)} = \frac{f'(c+h) - f'(c-h)}{2h} = \frac{1}{2}\Big(\frac{f'(c+h) - f'(c)}{h} + \frac{f'(c-h) - f'(c)}{-h}\Big)$$
$$\longrightarrow \frac{1}{2}(f''(c) + f''(c)) = f''(c).$$

Step 3. By L'Hôpital's rule (Theorem 4.9), $N(h)/D(h) \to f''(c)$.

(b) $f(x) = x|x|$, $c = 0$: $N(h) = 0$, so the limit is 0; but $f'(x) = 2|x|$ (at 0: $f(h)/h = |h| \to 0$) is not differentiable at 0. ■